

Surface chemistry and electrolyte concentration govern early-stage SEI formation on graphite anodes

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ABSTRACT

The solid electrolyte interphase (SEI) is crucial for the performance and safety of lithium-ion batteries. In conventional systems, the SEI results from the reductive decomposition of electrolyte molecules upon contact with the graphite anode surface. While SEI formation is strongly influenced by graphite surface chemistry, the specific role of naturally occurring oxygen-containing functional groups remains poorly understood. We bridge this gap by combining reactive force field molecular dynamics (ReaxFF-MD) and an implicit electrochemical potential to atomistically simulate the initial stages of SEI development under realistic charge-discharge conditions. We consider $-\text{CHO}$, $-\text{COOH}$, $-\text{COH}$, and $-\text{CO}$ functionalized graphite anodes and ethylene carbonate (EC)/dimethyl carbonate (DMC) electrolyte mixtures (1: 1 vol) at several LiPF_6 salt concentrations (1, 2, and 4 M). Our results indicate that surface chemistry determines solvent decomposition pathways and product distribution: $-\text{OH}$ groups stabilize EC and DMC, whereas accelerates decomposition and gas evolution. Functional groups $-\text{CHO}$, $-\text{COH}$ promote inorganic products like LiF , while groups $-\text{COOH}$, $-\text{CO}$ favor organic species. Increasing salt concentration enhances solvent decomposition, gas formation, and inorganic content, improving Li^+ transport but also increasing the polymerization of organic molecules. An AI-powered comparative analysis conducted on 24 high-impact studies on the subject confirms that our work fills three key gaps in the state-of-the-art in the simulation of SEI formation: being the first systematic MD study of four oxygen-containing functional groups on graphite anodes, it pioneers cyclic charge-discharge simulations to capture dynamic SEI evolution in Li-ion batteries and unravels the multi-dimensional effect of salt concentration on solvent decomposition and SEI structure. Overall, our results elucidate the versatile role of graphite surface chemistry and salt concentration in solvent decomposition and SEI nucleation and highlight the potential for tailoring surface functionalization to optimize SEI composition and enhance interfacial stability in lithium-ion batteries.

1. Introduction

The solid electrolyte interphase (SEI) is widely recognized as one of the most critical yet least understood components of lithium-ion batteries (LIBs). Formed in situ during the initial charging cycles through electrolyte reduction at the negative electrode, the SEI affects both performance and safety by passivating the electrode surface while permitting Li-ion transport [1,2]. Despite decades of research, its heterogeneous composition, nanometric thickness, and dynamic evolution

under cycling continue to pose major challenges for experimental characterization. Conventional surface-sensitive techniques such as X-ray Photoelectron Spectroscopy (XPS), Transmission Electron Microscopy (TEM), and Infrared (IR) Spectroscopy provide valuable insights into SEI composition and morphology, yet they struggle to capture the transient intermediates and ultrafast reactions that dominate the earliest stages of interphase formation [3,4].

Computational modelling has therefore become indispensable for elucidating SEI chemistry across multiple scales. At the atomistic level,

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density functional theory (DFT) and ab initio molecular dynamics (AIMD) have clarified the primary reduction pathways of carbonate solvents such as ethylene carbonate (EC) and dimethyl carbonate (DMC), including ring-opening, radical recombination, and the formation of inorganic salts such as Li_2O , Li_2CO_3 , and LiF [5–7]. However, AIMD is inherently limited to small system sizes and picosecond time-scales, precluding the study of mesoscopic clustering and long-term SEI growth. To overcome these constraints, reactive force field MD (ReaxFF-MD) has been employed to simulate electrolyte decomposition and SEI build-up on lithium metal, capturing multilayer structures and gas evolution in temperature-accelerated dynamics consistent with experimental observations [7–11]. At even larger scales, kinetic Monte Carlo (kMC) and hybrid MC–MD approaches have extended the accessible time and length scales, reproducing layered SEI architectures and quantifying species distributions over nanometers and microseconds [5,12–14].

While these advances have significantly deepened our understanding of SEI formation, reactive MD modelling efforts have focused on idealized graphite surfaces, often neglecting the chemical heterogeneity of practical anodes. Graphite electrodes are naturally enriched with oxygen-containing functional groups (FGs) such as hydroxyl, carbonyl, and carboxyl moieties, which strongly influence electrolyte adsorption, electron transfer, and decomposition pathways [15,16]. First-principles studies have demonstrated that such FGs can lower reaction barriers, alter product distributions, and even promote competing decomposition routes between EC, DMC, and fluoroethylene carbonate (FEC) [6,17]. Yet, despite these insights, systematic molecular investigations of how surface functionalization modulates SEI initiation under charge-discharge conditions remain scarce. In particular, classical ReaxFF-MD

simulations, highly successful in capturing SEI growth on metal anodes [9,17,18], have not been deployed to probe the influence of graphite surface chemistry on solvent decomposition and SEI evolution in the presence of oxygen-rich FGs.

This gap is significant, as FGs not only determine the initial adsorption and reduction of electrolyte molecules but also influence the balance between inorganic and organic SEI components, gas evolution, and interphase stability over cycling [6,17]. Moreover, existing state-of-the-art molecular modelling of the SEI tends to face three limitations: firstly, they often focus on one or two functional group modifications (e.g., $-\text{COH}/-\text{CO}$ [19]) at a fixed salt concentration (e.g., 1 M LiPF_6 [19]), failing to capture electrode – electrolyte synergistic effects. Secondly, static voltage simulations [20] over short timescales (< 10 ps in AIMD simulations [17]) are common and miss the reversible behavior of Li^+ in practical LIBs. Lastly, only the salt-dependent SEI thickness [21] or decomposition rate [16] is reported, without insight into how salt concentration regulates solvent decomposition pathways and SEI structural stability.

To overcome these limitations, we use ReaxFF-MD to investigate the initiation of SEI formation on a graphite anode similar to those in commercial LIBs, and an electrolyte mixture comprising EC/DMC (1,1 vol) and LiPF_6 salt. The electrochemical potential was set up using the Electrochemical Dynamics with Implicit Degrees of freedom (Echem-DID) method [22]. To understand the influence of oxygen-rich FGs, we decorated the graphite surface with ketonic ($-\text{CO}$), hydroxyl ($-\text{C}-\text{OH}$), carbonyl ($-\text{C}(=\text{O})-\text{H}$), and carboxyl ($-\text{C}(=\text{O})-\text{OH}$) terminations. For the ketonic group, which together with $-\text{COH}$ is predicted to be the most abundant edge group in graphite [15], we varied the salt concentration (1, 2, and 4 M LiPF_6) and evaluated the evolution and

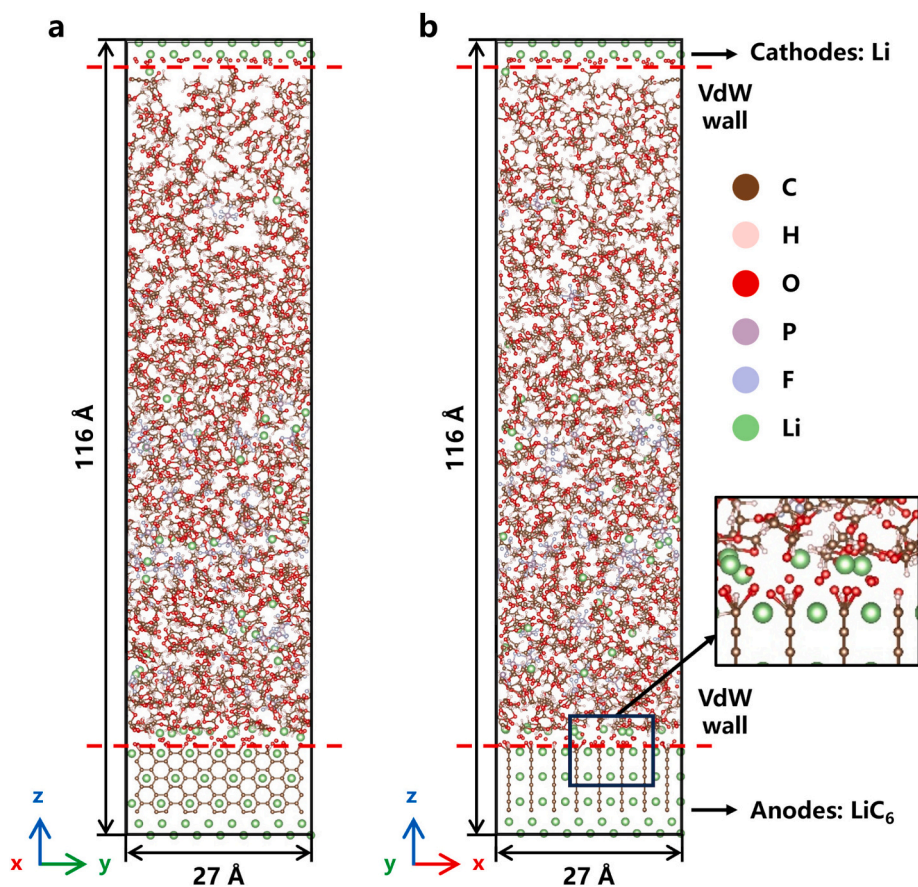


Fig. 1. Schematic diagram of model construction with a $-\text{CO}$ functionalized anode (bottom) and a metal Li cathode (top). The space between the electrodes is filled with an electrolyte solution consisting of LiPF_6 in a mixture of EC/DMC molecules at a 1:1 ratio. **a)** The front (yz -plane) and **b)** side (xz -plane) views of the initial configuration. The inset to the right depicts the structure of the anode surface.

distribution of solvent decomposition products during cycling.

2. Computational methods

2.1. Atomistic model

Fig. 1 shows the simulation model for this study. The upper part of the model represents a hypothetical lithium metal positive electrode (cathode), and the lower part is lithium-embedded graphite (LiC_6). Since we are exclusively interested in examining electrolyte decomposition at the functionalized graphite anode, the lithium cathode serves solely to maintain the voltage applied via the EChemDID algorithm. Thus, lithium atoms at the cathode are fixed, and no plating is permitted. Similarly, the lithium atoms within the graphite anode are immobilized to prevent exchange with the electrolyte phase, serving only to sustain the applied voltage. In this regard, alternating the electrodes polarity does not trigger intercalation – de-intercalation events, which would significantly increase computational complexity while not necessarily offering additional insights regarding the surface decomposition of electrolyte components. Pseudocathode/electrolyte interactions take place through a soft repulsive van der Waals (vdW) wall, which diffusely reflects incoming electrolyte molecules.

Hydrogen atoms are commonly observed near oxygen-rich FGs in graphite anodes, contributing to partial passivation of the graphite edges. Consequently, the FGs were introduced on every other edge carbon atom, totaling 48 sites. Similarly to the cathode side, a vdW wall is introduced under the first row of edge graphite atoms to avoid Li intercalation. The space between electrodes was filled with $\text{LiPF}_6/\text{EC}/\text{DMC}$ electrolyte solutions in proportions of 44/296/234, 88/262/208, and 116/195/155, which correspond to LiPF_6 salt concentrations of 1, 2, and 4 M, respectively. The initial number of solvent molecules was adjusted based on the experimental electrolyte density ($\sim 1.2\text{--}1.3\text{ g cm}^{-3}$) and the volume between the electrodes to produce $\sim 1:1$ (v/v) EC/DMC mixtures [23].

The structural models are constructed following established literature protocols [15], and the interaction energetics are benchmarked against existing first-principles studies on the same oxygen-containing functional groups on graphite edges [6,16]. The ReaxFF force field itself has been previously validated against quantum chemistry data for the relevant C/H/O/Li/P/F chemical space [25–27]. Periodic boundary conditions were imposed in the lateral (x - y) dimensions to mimic infinite electrode surfaces. Atoms in the cathode and below the top carbon layer in LiC_6 were fixed to prevent drift.

2.2. ReaxFF-MD simulations

All reactive molecular dynamics (ReaxFF-MD) simulations in this study were performed using the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) package [24], leveraging the ReaxFF reactive force field to explicitly capture bond breaking and formation during electrolyte decomposition and SEI growth. The employed ReaxFF parameter set was based on previously published C/H/O/Li/P/F parameters adopted from Castro-Marciano et al. [25] and Bedrov et al. [26], which were validated to reproduce quantum chemistry calculations of bond dissociation energies and Li-binding interactions for carbonate electrolytes and LiPF_6 salts [27], ensuring accurate description of EC/DMC solvent reduction, PF_6^- decomposition, and SEI products. All simulations were performed in the canonical (NVT) ensemble at 298 K using the Nosé-Hoover thermostat with a damping coefficient of 20 fs [28]. A time step of 0.2 fs was used to resolve fast bond-breaking events (e.g., EC ring opening, PF_6^- cleavage). Charge equilibration was performed on-the-fly using the shielded QEeq method [29,30] with a convergence criterion of 10^{-6} e, ensuring accurate calculation of atomic charges during SEI product formation (e.g., Li^+ coordination with F^- or carbonate anions).

Two distinct simulation workflows were implemented:

Graphite surface functional group (FG) simulations: to investigate the effect of graphite functionalization on SEI formation, we first simulated each FG-decorated LiC_6 anode at a 1 M LiPF_6 concentration. The systems were thermally equilibrated for 50 ps in the absence of external voltage to achieve stable interfacial adsorption and avoid voltage-driven reactions. Subsequently, 3 ns production simulations were performed at a fixed EchemDID voltage of 8 V to monitor solvent decomposition and SEI evolution.

Cycling simulations and LiPF_6 concentration effect: To understand the effect of lithium salt content on SEI product formation during cycling, we consider only the LiC_6 anodes modified with $-\text{CO}$. At each considered LiPF_6 concentration (1, 2, and 4 M), the system was first thermally equilibrated for 50 ps in the absence of external voltage. Charge-discharge processes were then simulated over 10 cycles, with each cycle consisting of 200 ps charge ($\Phi_0 = 8$ V) and 200 ps discharge ($\Phi_0 = -8$ V) stages established via EchemDID, totaling 4 ns of ReaxFF-MD simulations. Φ_0 represents the EchemDID voltage, which can be related to the overpotential through the method described by Intan et al. [17]. The EchemDID method [22,24] was employed to impose electrochemical potentials across the Li-metal cathode and FG-modified graphite anode, mimicking the voltage-driven reactions in practical LIBs. EchemDID modulates the atomic electronegativities $\chi_i^* = \chi_i^0 \pm \Phi_i/2$, of electrode atoms to establish a voltage difference, where Φ_i is the atomic electrochemical potential, dynamically updated via:

$$\Phi_i(t) = \sum_{j \neq i} \frac{\Phi_j(t) - \Phi_i(t)}{|R_{ij}|^2} \omega(R_{ij}) - \eta F(W_i) \Phi_i \quad (1)$$

Here, k denotes effective electrochemical potential diffusivity, R_{ij} is the interatomic distance, $\omega(R_{ij})$ is a local weighting function defining metallic clusters, η is the relaxation rate, $F(W_i)$ is a switching function that turns on the relaxation as an atom becomes detached from other metallic atoms, and W_i is the metallic coordination number. This formulation enables realistic simulation of electron transfer at the anode surface, driving reductive decomposition of EC/DMC and LiPF_6 without explicit inclusion of external current sources, consistent with previous ReaxFF studies of Li-metal anode cycling [9].

Detection of decomposition products was performed through the ReacNetGenerator software, a specialized graph-based tool designed for the automatic extraction of chemical reaction networks from reactive molecular dynamics simulation trajectories based on bond-order potentials [31,32]. Species concentrations c at a given time t are taken as an average of the instantaneous concentrations between $t - 50$ ps and $t + 50$ ps. To quantitatively monitor SEI nucleation, we adopt an operational definition based on this measurable criterion: the confinement of these fragments within a 5 Å cutoff from the top graphene layer (determined from atomic density profiles, Fig. S2), which distinguishes surface-bound nucleation species from bulk-decomposed products. SEI nucleation is considered to have commenced when all three criteria are met simultaneously. This definition ensures that our analysis captures the incipient interphase formation rather than general electrolyte decomposition.

For each condition, two independent ReaxFF-MD simulation runs were performed (3 ns for FG screening and 4 ns for cycling studies) to assess reproducibility. Comparison of the two runs showed excellent agreement in the relative reactivity order, concentration trends, and product distributions, with differences in final product concentrations within 0.02–0.05 M for organic species and 0.01–0.03 M for inorganic species. Given the computational cost and the consistent results between the two runs, the data presented are from one representative long trajectory, while the second run served as verification of repeatability. To further assess statistical stability, we performed a block-averaging analysis on the single trajectory.

3. Results and discussion

3.1. Effect of graphite surface functional groups on SEI formation

Figs. 2(a,b) present the variation in the concentration of EC and DMC molecules in the functionalized graphite anode models at 1 M LiPF₆ and an applied potential, Φ_0 , of +8 V, corresponding to an overpotential of +0.65 V (see Supporting Information). This overpotential corresponds to a high current rate and is therefore associated with a high decomposition scenario. $c_0 = 5.6$ M and 4.9 M refer to the concentrations of EC and DMC, respectively, after equilibration.

A progressive decline in the amount of both solvent molecules occurs over time. Fig. S1a,b in the Supporting Information shows that solvent decomposition is minimal in the bulk, indicating that decomposition is strongly promoted by the presence of the functionalized anode. Due to its cyclic carbonate structure, EC exhibits a stronger interaction with the polar oxygen-rich graphite surface than DMC, leading to a higher decomposition rate. These rates are highly dependent on the specific FG, with the hydroxyl group, -COH, exhibiting the highest degree of solvent stability: only 6% of EC and 2% of DMC are decomposed on -COH functionalized graphite over 3 ns. In contrast, the carboxyl group, -COOH, emerges as the most reactive FG, with decomposition rates of 20% and 19% for EC and DMC, respectively.

The edge surface of graphite is primarily populated by hydroxyl and ketonic (-CO) groups, due to their high thermodynamic stability. In contrast, carbonyl (-CHO) and carboxyl groups are significantly less favorable [15]. Thus, the actual solvent decomposition rate in a typical graphite anode is likely closer to an average between that of -CHO and -COH decorated surfaces (i.e., ~ 10% and ~ 5% for EC and DMC after 3 ns, respectively). The qualitative trends observed in our ReaxFF simulations are consistent with prior DFT adsorption energies [6] for the same functional groups, confirming that the force field captures the expected relative reactivity ordering without requiring additional reparameterization. Importantly, the high decomposition levels depicted in

Figs. 2(a,b) should be considered upper limits, given the large overpotential studied here and the high surface FG content. Nonetheless, the results indicate that graphite pre-treatment methods that modulate the surface FG ratios may contribute to mitigating undesired solvent loss.

The high solvent decomposition ratio in the -COOH model may result from the high exothermicity of solvent adsorption on this FG (Fig. 2(a,b)). According to the DFT-based analysis by Das et al. [6], the adsorption energy of DMC in the -COOH-decorated [11–20] graphite edge plane ranges between -5.6 and -4.3 eV, while the corresponding adsorption energy for -COH lies between -1.9 and -0.8 eV. Thus, although the decomposition energy of adsorbed DMC was found to be, on average, ~ 2 eV lower on -COH than on -COOH (depending on the specific decomposition route [6]), the overall adsorption + decomposition pathway is on average ~ 2 eV more exothermic on -COOH. Adsorption energies for EC on functionalized graphite surfaces have not been reported. However, DFT calculations by Rustam et al. [16] have shown that ethoxide-EC-Li complexes spontaneously adsorb on graphite across all considered surface functionalizations. Further investigation is therefore required to determine whether the complete adsorption + decomposition pathway of EC follows a trend similar to that observed for DMC.

Figs. 2(c-e) show the change in organic, inorganic and gaseous products in models with different FGs. These product classes are also produced by solvent decomposition in the bulk electrolyte (see Fig. S1d-e in the Supporting Information), albeit at considerably lower concentrations. The relative amounts of decomposition products vary with the anode functionalization, following the trends described below:

Organic products (Fig. 2c): -COOH leads to the highest concentration of these types of products (ROCO₂Li, LiO-R) after 3 ns. The rest of FGs yield comparable amounts of organic molecules, between 10 and 15% lower than -COOH functionalized graphite. Notably, the number of organic molecules is relatively stable after 1 ns, highlighting the high degree of activity that occurs immediately after the potential is established.

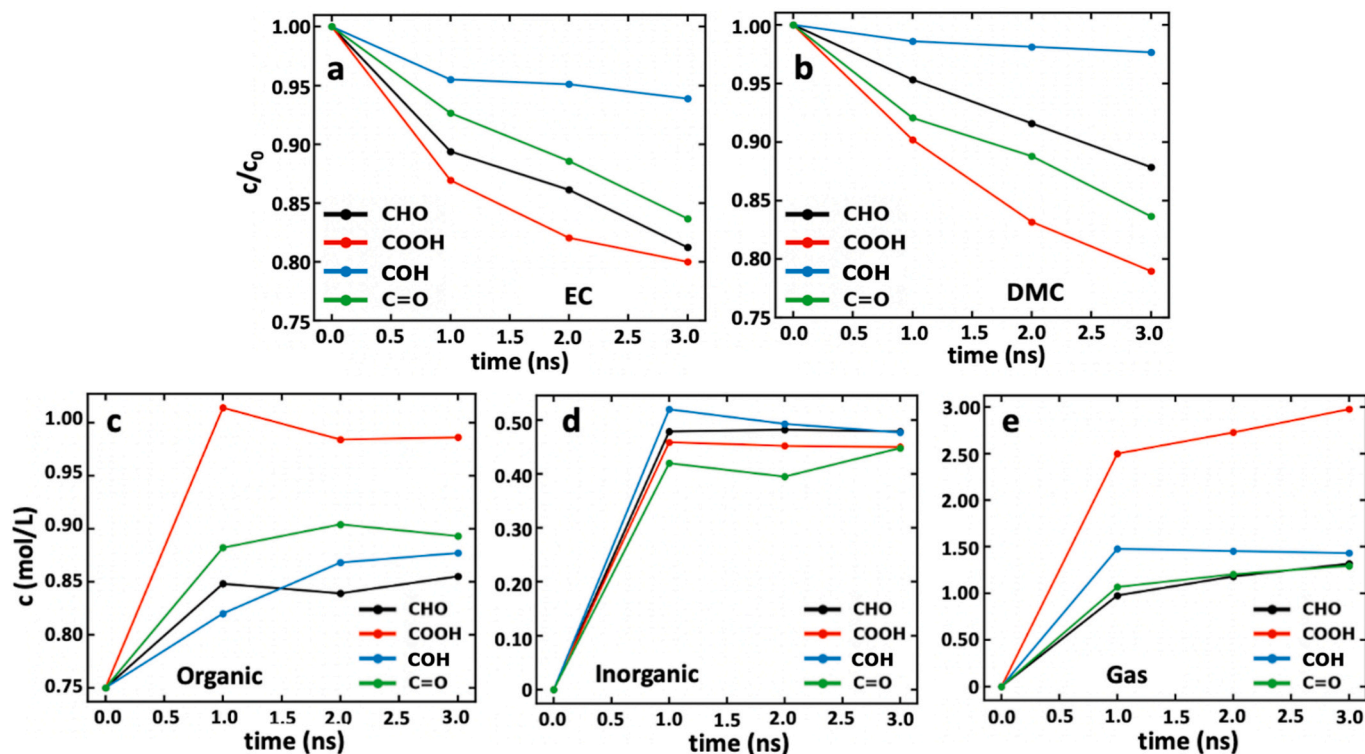


Fig. 2. Impact of graphite functionalization on solvent decomposition and product evolution at 1 M LiPF₆ and $\Phi_0 = 8$ V. (a,b) Fractional variation of solvent concentration (c/c_0) for a) EC ($c_0 = 5.6$ M) and b) DMC ($c_0 = 4.9$ M). (c-e) Variation in the concentration of product types: c) organics (e.g., LiO-R, LiO₂COR), d) inorganics (e.g., LiF, Li₂CO₃, LiOH), and e) gases (e.g., CO₂, C₂H₄).

Inorganic products (Fig. 2d): Interestingly, $-\text{OH}$ is, by a narrow margin, the strongest promoter of inorganic molecule formation, despite its minor role in solvent decomposition. The strong polarity of the $-\text{OH}$ group makes the surface electron density high, promoting Li^+ enrichment, PF_6^- decomposition, and, in consequence, the formation of inorganic products, such as LiF . The output of inorganic species stabilizes after 1 ns in all cases, and remains stable between 0.4 and 0.5 M. A considerable number of these compounds form early in the simulation, in agreement with previous reports suggesting the formation of a thin inorganic layer even before application of voltage [33,34].

Gas products (Fig. 2e): Solvent decomposition in the $-\text{COOH}$ model is more thorough, and gas products such as CO_2 and C_2H_4 are significantly more abundant than those produced from other FGs. Hydrogen atom transfer (HAT) processes play an important role in the reaction of $-\text{COOH}$ with DMC and EC [6,17], leading to the formation of a large number of gas molecules. In the $-\text{CO}$ model, DFT simulations suggest that the solvent largely undergoes partial decomposition (such as ring opening without complete bond breaking), and there are fewer gas products, due to the strong adsorption of decomposed radicals on the $-\text{CO}$ moiety [35]. Notably, the number of gas molecules increases monotonically for all FGs except for $-\text{COH}$, which corresponds precisely to the FG leading to the weaker solvent decomposition.

In summary, the oxygen-containing FGs anchored on the surface of graphite anodes exert a decisive regulatory effect on the solvent decomposition behavior, product composition, and reaction kinetics during the initial formation of the SEI. Among these functional groups, the $-\text{COH}$ group exhibits the most pronounced stabilizing effect on the EC and DMC solvents, as evidenced by the minimal reduction in solvent concentration observed within 3 ns. In contrast, the $-\text{COOH}$ group demonstrates the strongest catalytic activity towards solvent decomposition, achieving the highest cumulative decomposition ratio. Regarding product distribution, the $-\text{COOH}$ and $-\text{OH}$ groups are the most inclined to facilitate the evolution of inorganic SEI components such as LiF . Meanwhile, the $-\text{COOH}$ and $-\text{CO}$ groups tend to promote the formation of organic carbonates through solvent ring-opening polymerization. Notably, the $-\text{COOH}$ -containing system yields the most abundant gaseous products, especially CO_2 and C_2H_4 . These findings indicate that precise control over solvent stability and the organic/inorganic product ratio during SEI formation can be achieved by modulating the type of functional groups on the graphite surface. This route was recently explored by Xie et al. [36], who fine-tuned the fraction of O-containing FGs in graphene oxide anodes through thermal treatment.

3.2. Surface reactivity during charge-discharge cycles

In this section, we focus on the LiC_6 graphite anode modified with $-\text{CO}$, as this FG displays generally high reactivity (see Fig. 2) while also being among the most abundant and electrochemically active in Li-ion batteries [15]. Three lithium salt concentrations (1 M, 2 M, 4 M LiPF_6 in $\text{EC}/\text{DCM} = 1,1$ vol) were tested over 10 charge-discharge cycles, each cycle lasting 400 ps (200 ps charge and 200 ps of discharge with opposite voltage directions) for a total duration of 4 ns.

Fig. 3 illustrates that the decomposition of EC (Fig. 3a) and DMC (Fig. 3b) increases with increasing lithium salt concentration. This is explained by the enhanced coordination between Li^+ and EC/DMC molecules at high LiPF_6 contents, which reduces the electron cloud density of the solvent molecules and facilitates the acceptance of electrons from the electrode surface for reductive decomposition [23]. The corresponding density profiles of the solvent molecules along the z -axis are depicted in Fig. S2 of the Supporting Information, revealing a largely homogeneous distribution of EC and DMC across the electrolyte in all scenarios. That is, EC/DMC decomposition at the anode is a surface-controlled process, with diffusion rapidly replenishing any spent solvent.

Interestingly, a large decrease in the initial concentration of EC ($c_0 = 5.6$ M) and DMC ($c_0 = 4.9$ M) occurs at the onset of cycling (0th in the x -axis of Fig. 3 panels) for 2 and 4 M LiPF_6 (dashed blue and red lines, respectively). Thus, even before an external voltage is established (i.e., at $\Phi_0 = 0$), significant amounts of solvent may already be adsorbed and undergo decomposition at high salt contents. During this stage, the active sites on the $-\text{CO}$ functionalized graphite surface are fully exposed, and the large amount of dissolved Li^+ enhances Li-mediated adsorption and decomposition [6]. In addition, the two-electron decomposition paths of EC and DMC [6], which require simultaneous attack by two Li^+ ions, will be more frequently observed at high salt contents.

Once $\Phi_0 \neq 0$, decomposition is most significant during the first two cycles, particularly for 2 and 4 M LiPF_6 . Following the second cycle, the solvent consumption rates drop sharply and begin to stabilize. For instance, in the 2 M electrolyte, EC decomposition between the 2nd and 10th cycles accounts for only 6% of the total 28% decomposition observed during cycling. A similar trend is seen for DMC, where decomposition over the same interval contributes just 6% to the total 22% loss. This stabilization is attributed to the onset of Solid Electrolyte Interphase (SEI) formation, which partially passivates the active sites.

Seoung-Bum et al. [37] recently demonstrated that highly concentrated carbonate-based electrolytes (~ 5 M) can facilitate the formation

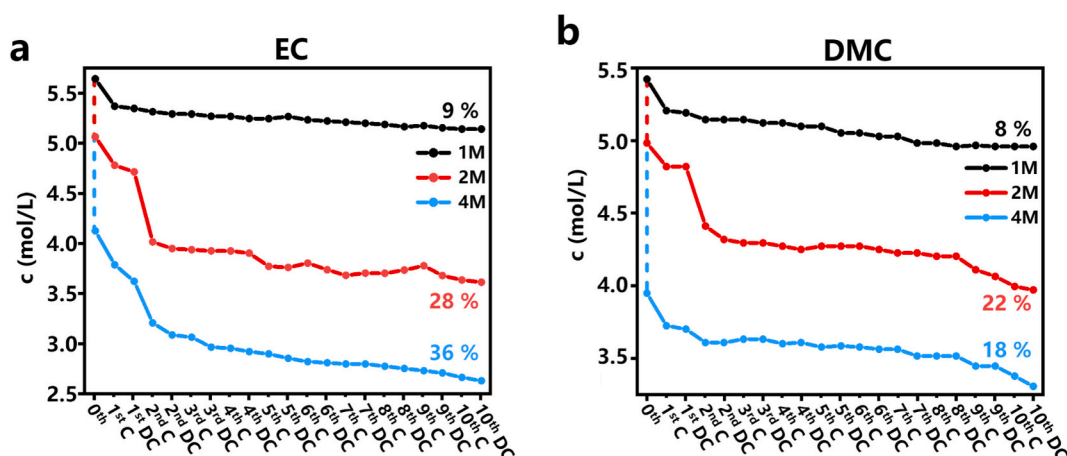


Fig. 3. Variation of solvent concentration with cycling in the $-\text{CO}$ functionalized graphite anode at different salt concentrations (1, 2, and 4 M LiPF_6) for a) EC ($c_0 = 5.6$ M), and b) DMC ($c_0 = 4.9$ M). Only the concentration at the end of each cycle is shown (except for the first cycle, for which the starting point, marked by 0th in the x -axis, is shown). The fractions of decomposed solvent during cycling are displayed at the end of each curve. Blue and red dashed lines show the drop of solvent concentration during equilibration at 2 and 4 M LiPF_6 . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of a reversible protective shielding layer on graphite anodes under an applied electric field. Unlike the permanent SEI found in dilute (~ 1 M) systems, this transient layer dissolves upon removal of the potential, effectively isolating the charged surface and suppressing electrolyte decomposition. While our current model, potentially limited by the absence of intercalation and short cycling durations, did not show a reduction in decomposition at high salt concentrations, we did observe solvent decomposition at open-circuit voltage. This suggests that significant solvent consumption may occur prior to the formation of the shielding layer. Furthermore, concentrations exceeding 4 M notably increase viscosity and hinder transport efficiency [38]. Consequently, for the electrolyte system studied here (without sacrificial additives), lower salt concentrations appear to be the more viable configuration.

Fig. 4 depicts the evolution of gas and organic products during cycling, while **Fig. 5** presents the evolution of inorganic products. Below, we describe the predicted trends in the formation of these product types at the examined salt concentrations under cycling.

Gas products: According to **Fig. 4a**, the amount of gas products (mainly CO_2 and C_2H_4) increases significantly at 2 and 4 M LiPF_6 , stabilizing at ~ 1.8 M – 1.9 M. This is likely due to enhanced electron transfer efficiency on the electrode surface, providing energy for decomposition of solvent molecules and the evolution of gaseous by-

products. Interestingly, gas evolution is slower at 4 M than 2 M, due to competition from other reaction pathways that are enhanced by increasing salt content (e.g., solvent polymerization, see discussion below).

The presence of CO_2 and C_2H_4 has previously been detected through differential electrochemical mass spectrometry (DEMS) in LiPF_6/EC :DMC electrolyte mixtures upon cycling [39]. Moreover, recent AIMD simulations of $-\text{CO}$ functionalized graphite have predicted the production of small quantities of CO_2 (but not C_2H_4) from DMC [6]. In our simulations, reactNetGenerator analysis identified C_2H_4 as originating exclusively from EC decomposition, in agreement with these results. In addition, **Figs. 4c-d** indicate that the amount of CO_2 generated by EC decomposition is significantly higher than that generated from DMC (**Figs. 4c-e**). Indeed, prior ReaxFF simulations of bulk LiPF_6/EC electrolytes have predicted the production of CO_2 and C_2H_4 from EC even under no external voltage [40] (we have reproduced this behavior and show it in **Fig. S1d** in the Supporting Information).

Analysis of reaction sites in the solvent molecules (**Fig. 4f**) indicates that EC can generate CO_2 through either unimolecular decomposition or bimolecular interaction, while DMC generates CO_2 through unimolecular decomposition only [5]. This fact, together with the higher reactivity of cyclic carbonates towards oxygen-containing functional

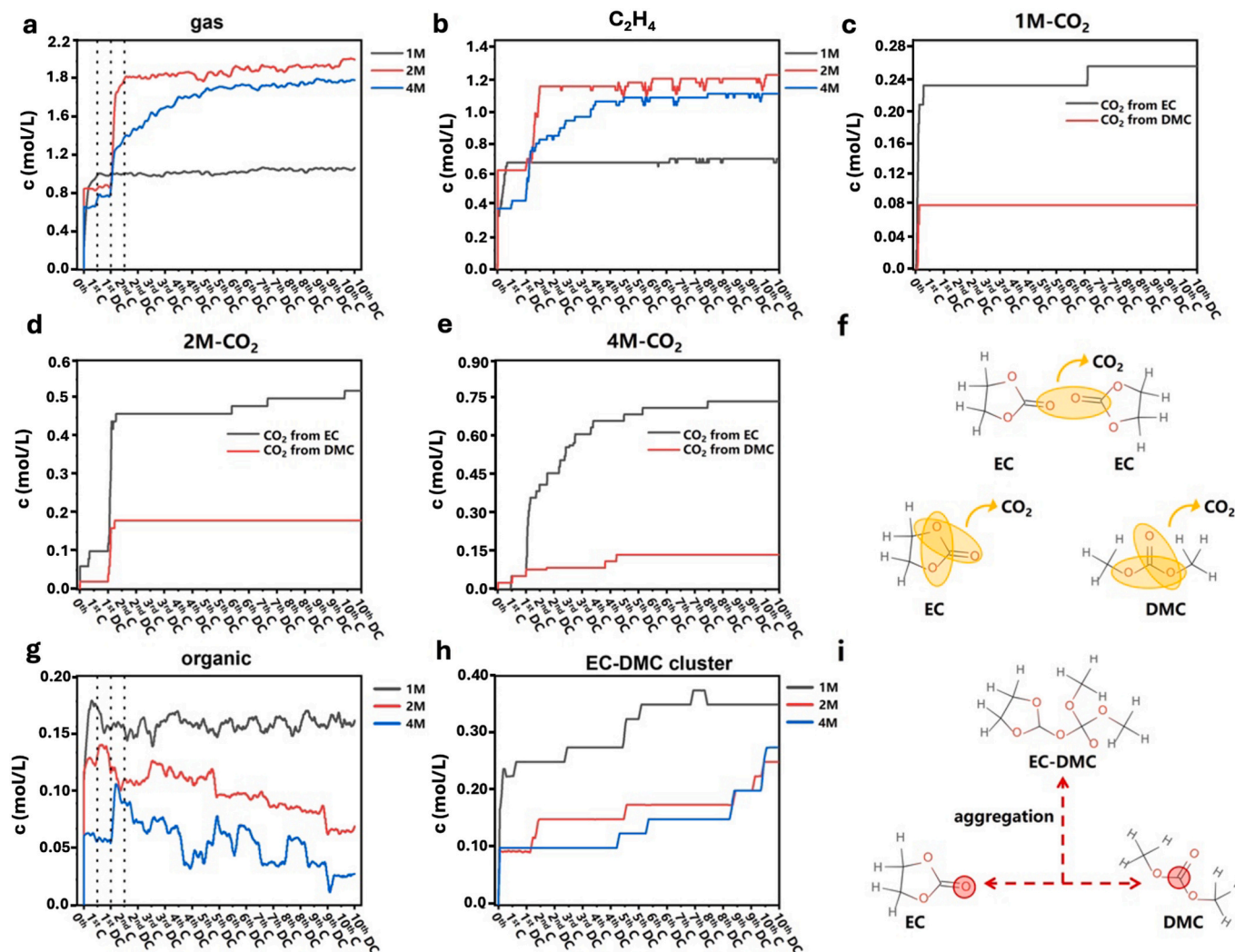


Fig. 4. Variation of organic and gas products during cycling in the $-\text{CO}$ functionalized graphite model at different LiPF_6 concentrations (1, 2, and 4 M). **a)** Changes in the number of gas molecules. **b)** Variation of C_2H_4 molecules. **c)-e)** Changes in CO_2 production from EC and DMC decomposition. **f)** The reaction sites where EC and DMC decompose to produce CO_2 . **g)** Variation in organic product molecules. **h)** The number of large clusters formed by the polymerization of EC and DMC. **i)** Schematic diagram of EC and DMC polymerization forming clusters.

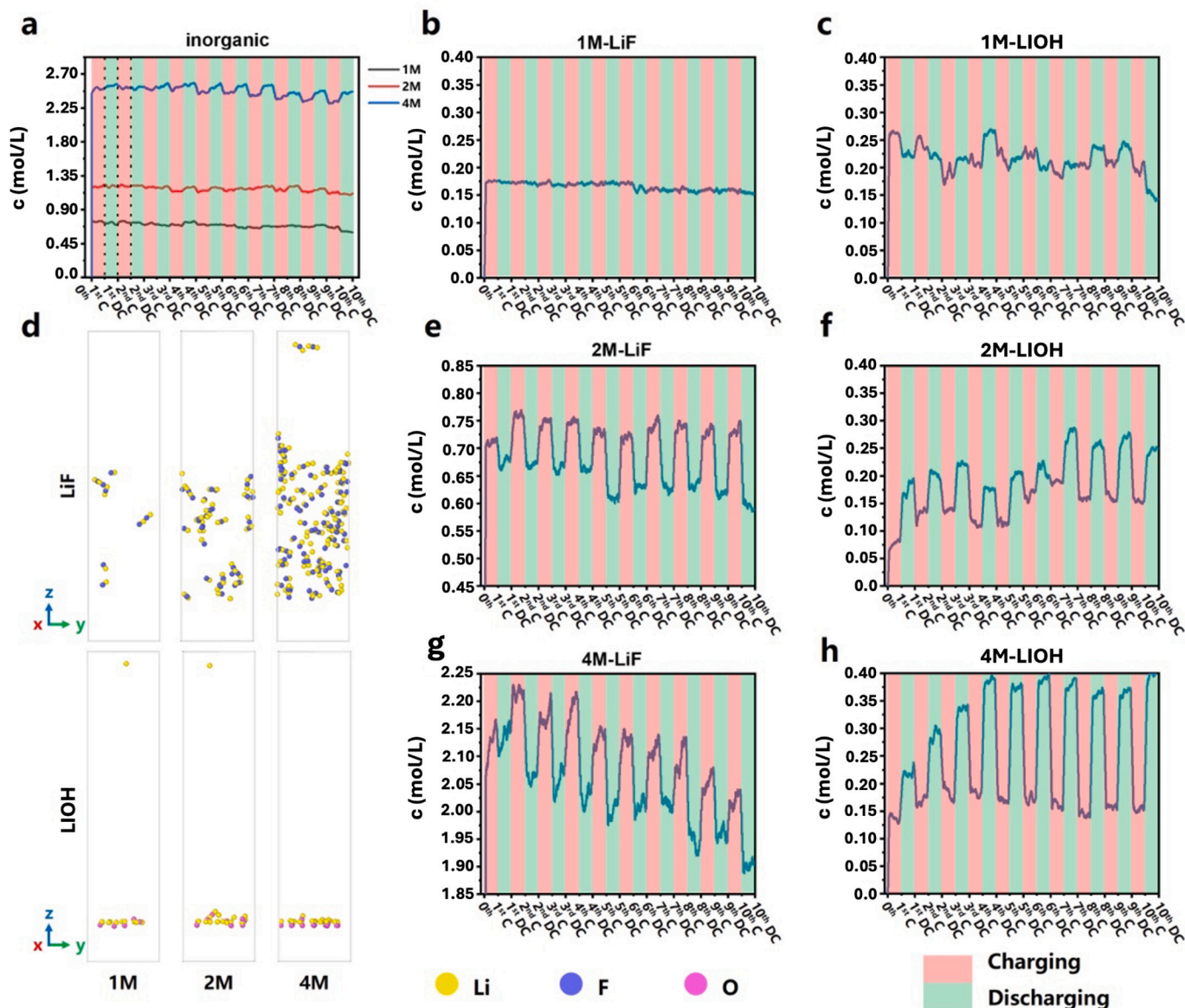


Fig. 5. Variation in the concentration of inorganic products during the cycling process in -CO functionalized graphite models with different salt concentrations (1, 2, and 4 M LiPF₆). **a)** All inorganic products. **b), e), and g)** Changes in the quantity of LiF during cycling in different lithium salt concentration models. **c), f), and h)** Changes in the quantity of LiOH during cycling in different lithium salt concentration models. **d)** Distribution of LiF and LiOH in models with different lithium salt concentrations after the tenth discharge cycle.

groups, explains the greater production of CO₂ from EC with respect to DMC.

Organic products: Fig. 4g shows that the concentration of organic products decreases slightly with the increase of lithium salt concentration at 2 and 4 M and fluctuates during the cycle. At 1 M, it remains relatively constant. A decreasing number of organic molecules does not necessarily stem from a reduction in the production of organic compounds, but results from increasing polymerization, which leads to longer chains at the expense of smaller organic fragments. To test this hypothesis, Fig. 4h shows that the number of clusters formed by the polymerization of EC and DMC increases with the cycle (see schematic diagram shown in Fig. 4i). The number of such clusters decreases with salt concentration, indicating the formation of longer but fewer organic molecules.

Limited polymerization often stabilizes the SEI and improves ion transport, while excessive polymerization can increase ionic impedance and reduce battery performance [41,42]. Our work suggests that the

most typical salt content (1 M LiPF₆) leads to stable polymer chain lengths, but of smaller size than those produced at 2 and 4 M LiPF₆. Hence, while ~1 M LiPF₆ is often chosen because it provides maximum electrolyte conductivity and solubility [43], our simulations suggest this salt concentration may also benefit the formation of the most stable polymer substrate for controlled SEI growth.

Inorganic products: According to Fig. 5a, the number of inorganic molecules remains relatively stable after the first cycle, with increasing salt concentrations naturally leading to higher inorganic product contents. The main inorganics predicted by our model are LiF (Figs. 5b,e,g) and LiOH (Figs. 5c,f,h), which, along with Li₂CO₃ (Fig. S3 of the SI) and Li₂O (detected in trace amounts), are the key inorganic components of the SEI [1–3]. Interestingly, LiF and LiOH display opposing evolution trends: during charge, the amount of LiF increases while that of LiOH decreases, whereas the reverse occurs during discharge. This reciprocal behavior suggests a dynamic interconversion of Li⁺ species at the anode surface (where LiOH is adsorbed, see Fig. 5d); the extent of these

fluctuations becomes increasingly pronounced with rising salt concentration. We also note that the concentration of LiF increases by a factor of ~ 10 between 1 M and 4 M LiPF₆ after the 10th cycle, while the concentration of LiOH increases by a factor of ~ 2 . This is likely due to LiF forming upon reduction of LiPF₆ at the anode surface during charging. However, this species can subsequently diffuse towards the

bulk electrolyte (see Fig. 5d (top)) and release active surface sites. On the contrary, LiOH is surface-bound, and its concentration is limited to the availability of surface sites.

Notably, the concentration of LiF exhibits a progressive decline with cycle number at 4 M LiPF₆, while the concentration of LiOH remains stable. This indicates that the high salt concentration leads to rapid LiF

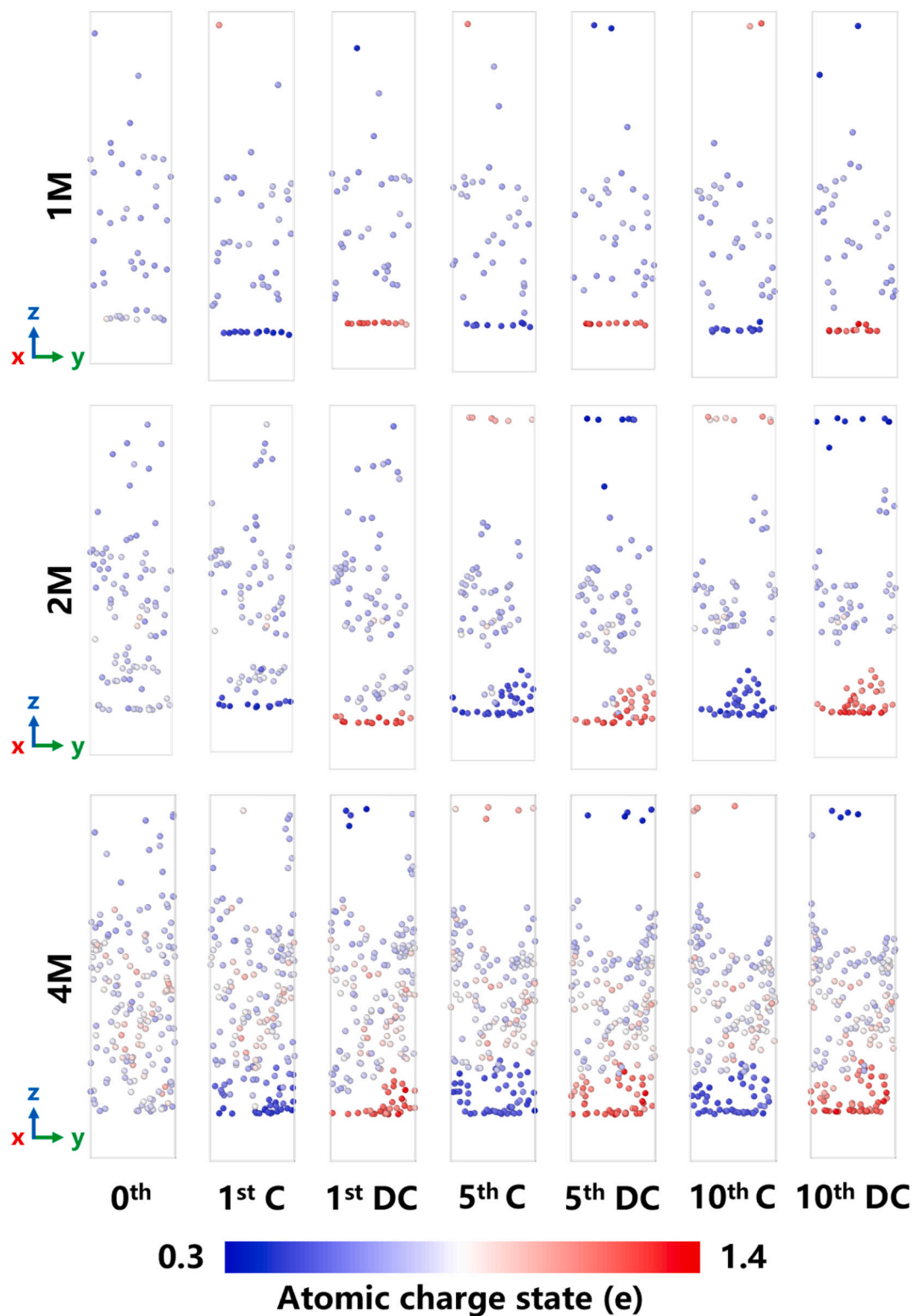


Fig. 6. Charge distribution of all Li in models with lithium salt concentrations of 1, 2 and 4 M LiPF₆ during the cyclic charge-discharge process. The color bar indicates the atomic charge state.

formation during the first cycles, depleting the available F and hindering LiF replenishment during the interconversion process.

As a critical inorganic precursor in the SEI, lithium hydroxide (LiOH) reacts with carbonates to produce Li_2CO_3 and undergoes the lithiation reaction $\text{LiOH} + \text{Li}^+ + \text{e}^- \rightarrow \text{Li}_2\text{O} + \text{LiH}$. However, an excess of LiOH leads to SEI instability, continuous growth, and rapid capacity decay in subsequent cycles [44]. Our observations show a decrease in LiF concentration during cycling at 4 M LiPF₆, alongside a relatively high and stable concentration of LiOH. This indicates that such high salt concentrations, without additional additives, are prejudicial to the formation of a stable and optimally performing SEI [45].

Li_2CO_3 , another critical SEI component, is predicted to form only in small quantities (see Fig. S3 in the SI). Since LiOH is one of its known precursors [44], we hypothesize that the reaction of surface-bound LiOH and EC/DMC to produce Li_2CO_3 is kinetically hindered under the specific limitations of our model, such as the absence of intercalation and short cycle duration. Nevertheless, the formation of these species indicates that the key chemical processes necessary for SEI growth are present in our model, and suggests that further improvements may be possible to accurately simulate the nucleation and densification of the inner SEI layer.

In Fig. 6, the charge distribution of lithium at the end of each cycle shows significant variation with salt concentration. At high concentrations, the charge state of lithium ions is more stable (with smaller fluctuations). This phenomenon is consistent with the interpretation that increased inorganic product content (such as LiF) in the SEI enhances the ionic conductivity of the interface, making the migration of lithium ions (Li^+) smoother, which is of great significance for inhibiting dendrite growth. Importantly, the charge state of the ions is most considerably changed near the electrode surfaces, because reactivity is enhanced by the state of charge [9].

A schematic of the atomic distributions of C, H, O, and solvent decomposition residues for salt concentrations 1, 2, and 4 M LiPF₆ is shown in Figs. S4, S5, and S6 of the SI, respectively. In general, a largely homogenous elemental distribution is observed, showing that the decomposition kinetics does not introduce additional diffusion barriers.

4. Conclusion

Our study investigates the regulatory effects of oxygen-containing functional groups (FGs) on graphite anodes and LiPF₆ concentrations on early-stage solid electrolyte interphase (SEI) formation via reactive force field molecular dynamics (ReaxFF-MD) simulations, coupled with the EchemDID method for electrochemical potential modulation.

The research findings indicate that the type of functional groups on the LiC_6 surface has a significant influence on solvent decomposition behavior and the composition of SEI products. A substantial fraction of solvent decomposition is observed at an EchemDID voltage of $\Phi_0 = 0$ V, corresponding to the equilibration stage. This likely happens because, under $\Phi_0 = 0$ V, molecular transport is dominated by tracer diffusion rather than being driven by an electrochemical potential gradient, enabling solvent molecules near the surface to interact with available functional groups. It is important to note that, in larger-scale devices, decomposition at $\Phi_0 = 0$ V would be reduced due to diffusion limitations; therefore, the present results should be regarded as an upper bound for solvent decomposition in an environment with a large excess of FGs.

Among the four functional groups examined (–CHO, –COOH, –COH, –CO), the –OH group demonstrates the strongest ability to stabilize solvents (EC and DMC), while the –COOH group, due to its highly polar O–H bond and the presence of a –CO termination upon –COH detachment, accelerates solvent decomposition by enhancing the adsorption of carbonyl groups in solvents through hydrogen bonding.

In terms of SEI products, –CHO and –COH promote the formation of inorganic products (such as LiF) by facilitating Li^+ enrichment and PF₆[–] decomposition, whereas –COOH and –CO favor the evolution of organic

products (like organic carbonates) through ring-opening polymerization of solvents. Additionally, the –COOH model exhibits more thorough solvent decomposition, resulting in more gas products (e.g., CO_2 , C_2H_4), while the CO model produces fewer gas products due to partial solvent decomposition. Interestingly, large quantities of –COOH have been associated with parasitic, gas-generating reactions. However, it has been reported to be beneficial in small amounts. Our model clearly contains a large excess of –COOH, leading to a larger-than-usual gas concentration.

During charge-discharge cycles, increasing the lithium salt (LiPF₆) concentration (from 1 M to 4 M) significantly accelerates the decomposition of EC and DMC, which is attributed to enhanced coordination between Li^+ and solvent molecules (reducing solvent electron cloud density) and the strengthened interaction between PF₆[–] and Li^+ (promoting solvent diffusion to the electrode surface). High concentrations also lead to more gas products (primarily from deeper solvent decomposition) and a slight decrease in organic products, along with an increase in the number of large clusters formed by EC/DMC polymerization (larger polymer chains).

For inorganic products, LiF and LiOH show opposite quantity trends during charging and discharging (reflecting Li^+ interconversion), with higher concentrations causing more significant fluctuations and increased LiF content (due to sufficient Li^+ sources). Furthermore, higher concentrations stabilize the charge state of Li^+ by increasing inorganic products in the SEI, improving interfacial ionic conductivity and potentially inhibiting dendrite growth. These results provide insights into regulating SEI formation in lithium-ion batteries by adjusting graphite surface chemistry and electrolyte concentration, which is beneficial for optimizing battery performance.

The AI-powered comparative analysis validated three core innovations that address SOTA limitations. First, it resolves SOTA's "single-FG limitation": SOTA studies (e.g., [19]) typically focus on single-FG modification or no FGs at all, failing to clarify how FG polarity regulates SEI products, while our systematic simulation of four FGs establishes a quantitative "FG polarity-SEI composition" correlation that fills this gap. Second, cyclic charge-discharge simulations (10 cycles, 4 ns total) capture dynamic SEI evolution, revealing novel phenomena such as LiF-LiOH interconversion and SEI passivation saturation (70% reduction in solvent decomposition rate after 2 cycles in 1 M systems)—breaking through SOTA's static simulation constraints. Third, this study decipheres the multi-dimensional role of salt concentration, clarifying its regulation of solvent decomposition pathways and SEI compactness (high concentrations disrupt ordered SEI arrangement, causing sustained low-rate decomposition)—bridging the gap between SOTA's macroscale structural observations and atomic-scale reaction mechanisms.

Collectively, this study demonstrates that tailoring graphite surface chemistry (FG type and polarity) and electrolyte composition (LiPF₆ concentration) is a viable strategy to optimize SEI composition, balance organic/inorganic phases, and enhance interfacial stability. The integrated findings provide actionable guidelines for designing high-performance LIB anodes, offering a foundation for improving battery safety and long-term cycling performance. The reproducibility of the simulation results was confirmed by two independent runs per condition, which showed consistent trends. Block-averaging analysis further confirmed the statistical stability of the reported trends. Future work incorporating additional repeated trajectories would further enhance the quantitative reliability of the predictions.

Author contribution

Menghao Yang conceived and supervised the project. Henry A. Cortes and Yilin Chen designed the computation and analysis. Yilin Chen, Mauricio R. Bonilla, Bikram K. Das and Elena Akhmatskaya wrote the manuscript. All authors participated in the discussion and interpretation of the data.

CRediT authorship contribution statement

Menghao Yang: Conceptualization. **Yilin Chen:** Formal analysis. **Mauricio R. Bonilla:** Project administration. **Henry A. Cortes:** Investigation. **Bikram K. Das:** Investigation. **Elena Akhmatkaya:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.commatsci.2026.115056>.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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